

Docker Image Versioning Strategy (Explanation + Demonstration)

1. Introduction: Docker image versioning means assigning tags (like v1, v2) to manage different versions of an application.

2. Why Versioning is Important: - Helps rollback to previous versions - Avoids confusion with latest tag - Enables proper deployment tracking

3. Demonstration Steps:

Step 1: Create app.py

```
print("Version 1")
```

Step 2: Create Dockerfile

```
FROM python:3.10
COPY app.py /app.py
CMD ["python", "/app.py"]
```

Step 3: Build Version 1

```
docker build -t myapp:v1 .
```

Run Version 1

```
docker run myapp:v1
```

Step 4: Modify app.py to Version 2

```
print("Version 2")
```

Build Version 2

```
docker build -t myapp:v2 .
```

Run Version 2

```
docker run myapp:v2
```

Step 5: List Images

```
docker images
```

Step 6: Rollback Example

```
docker run myapp:v1
```

4. Best Practices: - Use semantic versioning (v1.0, v2.0) - Avoid only using latest - Push tagged images to Docker Hub

5. Scenario Question: If version v2 has a bug, rollback using v1 and fix code to release v3.